

MUSHARRAF AZIZ

AI Engineer | Applied AI and Production Grade Machine Learning

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PROFILE

Applied AI Engineer with more than five years of experience designing, building, and deploying AI and machine learning systems into live production environments, not experimental or research settings. Direct experience across large language models, retrieval augmented generation, AI agents, predictive modelling, and computer vision, shipped into healthcare, e commerce, and energy businesses. Comfortable owning a system end to end, from data pipelines and model development through evaluation, deployment, and monitoring on Azure and AWS. Background also includes part time work training others in applied AI, and coordinating emergency response efforts as a volunteer, both of which have sharpened the ability to explain technical work clearly and operate under pressure with limited resources.

WORK EXPERIENCE

Senior Applied AI and Machine Learning Engineer

Cygnus Technologies

Lahore, Pakistan | Full time | July 2026 to Present

- Design, build, and deploy AI and machine learning solutions into production environments, covering large scale data processing, predictive analytics, and intelligent automation.
- Build data pipelines and integration architectures for high volume data ingestion, transformation, and structured storage, aimed at improved performance and reduced latency.
- Develop and fine tune large language model and retrieval augmented generation applications using LangChain and LlamaIndex with vector database storage, alongside Hugging Face and OpenAI APIs.
- Deploy, monitor, and optimise production machine learning models on AWS following MLOps practices, covering versioning, performance monitoring, and reliable rollout.
- Extend PII redaction and data governance controls using spaCy named entity recognition alongside Microsoft Presidio across enterprise data pipelines.
- Build evaluation and hallucination detection tooling for internal LLM applications, scoring answer relevancy and faithfulness against golden datasets and tracking pass rate and latency over time.
- Collaborate with cross functional stakeholders to translate business needs into technical specifications for AI models integrated into production systems.

Trainer, Applied Artificial Intelligence

Bano Qabil Pakistan (Alkhidmat Foundation initiative)

Kasur, Punjab, Pakistan | Part time, Volunteership | August 2026 to Present

- Deliver hands on training in applied artificial intelligence, machine learning, deep learning, and generative AI to learners with little or no technical background.
- Design and deliver a curriculum focused on real world AI implementation and end to end AI application development, built around practical project work.
- Teach Python, large language models, prompt engineering, retrieval augmented generation, AI agents, embeddings, vector databases, and modern AI frameworks in plain, approachable language.
- Mentor and assess learners through hands on projects, and cover AI deployment, MLOps fundamentals, responsible AI, safety, ethics, and governance.

AI Engineer and Operations Manager

Allama Iqbal Hospital, Kasur

Kasur, Pakistan | Full time | August 2024 to July 2026

- Designed and deployed an agentic AI CallBot on LLaMA 3.3 70B served through Groq, with OpenAI and Gemini as additional providers, reachable over WhatsApp and inbound phone calls through Twilio, handling more than 1,000 daily interactions in production.

- Built and optimised a retrieval augmented generation pipeline in LangChain, grounding agent responses in operational knowledge and applying prompt engineering to reduce hallucinated or incorrect answers.
- Built agent workflows connecting the AI system to internal databases, REST APIs, and messaging platforms through a 16 node automation pipeline in n8n, and built a Model Context Protocol server giving agents structured, tool based access to external systems.
- Monitored agent output and decision quality over time, reviewing real interactions to identify failure patterns, building lightweight evaluation checks, and refining retrieval logic and prompts to improve reliability.
- Hardened the agent against low confidence or incorrect outputs using fallback handling and data governance controls, including PII detection and redaction with spaCy and Microsoft Presidio, applied before any data reached the language model.
- Built PyTorch based LSTM neural networks for patient flow forecasting, improving scheduling accuracy, and GAN based models for structured data and text to image generation.
- Deployed and hosted AI services on Microsoft Azure for close to two years, covering compute, storage, and model hosting for the hospital's AI platform.
- Built OCR based document processing using Tesseract and PaddleOCR open source libraries in Python, extracting structured data from scanned documents and images for downstream automation.
- Recipient of the High Performance Excellence Award, June 2025.

Automation Engineer and IT Manager

NovaSole Pakistan

Kasur, Pakistan | Full time | December 2023 to August 2024

- Built data pipelines and automated synchronisation logic across three sales channels for an e commerce platform serving more than 500,000 monthly visitors, achieving more than 98 percent data accuracy.
- Built automated workflows connecting the platform to payment processors and inventory systems using REST APIs and webhook triggers.

Team Lead, Quality Assurance and Network Operations Centre

Ihsan Solar Energy Pvt. Ltd.

Raiwind, Pakistan | Full time | December 2022 to December 2023

- Built monitoring and alerting logic for a Network Operations Centre, using operational data to identify patterns and improve fault detection across more than 400 kilowatts of installed capacity.
- Led a quality assurance team of four, introducing structured testing and data driven analysis that reduced operational faults by 25 percent.

Team Lead, Technical Assistance Centre

Transworld Home (Internet Service Provider)

Lahore, Pakistan | Full time | March 2022 to November 2022

- Led a team of 14 engineers, using performance data to identify recurring issues, achieving 98 percent issue resolution within service level agreement across more than 50,000 active connections.

SELECTED AI ENGINEERING PROJECTS

- Agentic AI CallBot, WhatsApp and Twilio Voice: Designed and deployed an agentic conversational system on LLaMA 3.3 70B served through Groq, with OpenAI and Gemini as additional providers, backed by a LangChain retrieval augmented generation pipeline, connected to internal databases and messaging platforms through n8n workflows, handling more than 1,000 daily interactions in production.
- Git Archaeologist MCP Server: Built and published a Model Context Protocol server giving AI agents structured, tool based access to git history, resolving commits to pull requests through the GitHub API.
- Patient Flow Forecasting Model: Built and deployed a PyTorch based LSTM model for time series forecasting into a live production inference pipeline, including data preprocessing and ongoing monitoring for accuracy.
- AegisFlow Predictive Financial Models: Built PyTorch based LSTM neural networks for cash flow forecasting and GAN based generative models, including text to image generation, for financial stress testing, integrated into a FastAPI backed application with PostgreSQL storage.
- RAG Knowledge Base System: Built a multi source document ingestion pipeline with Hugging Face embeddings and ChromaDB and Pinecone vector storage, using LangChain retrieval chains for grounded question answering.

- LLM Evaluation and CI/CD Pipeline: Built an automated evaluation pipeline for LLM applications using Groq hosted models and DeepEval metrics, continuously testing responses against a golden dataset for hallucination detection, answer relevancy, and faithfulness, with results tracked through GitHub Actions and a live dashboard.
- LLM Guardrail Gateway: Built an asynchronous middleware layer sitting between client applications and upstream language models, applying inbound prompt sanitisation and jailbreak detection, enforced JSON schema validation on outbound responses, and asynchronous compliance audit logging.
- Waste Management Detection System: Built a computer vision pipeline for waste detection, classification, and segregation, applying object detection and classification models to categorise waste types for automated sorting.

EDUCATION AND TRAINING

B.S. (Honours) in Electrical Engineering

COMSATS University Islamabad, Lahore Campus

September 2017 to August 2021

- Final year project on a LoRaWAN based smart agriculture decision support system, awarded Top Innovative Final Year Project of the Year by IGNITE and the Higher Education Commission of Pakistan, 2021.
- Published as co author in Sustainability (MDPI), a peer reviewed journal with an impact factor of 3.125.

CERTIFICATIONS AND FURTHER TRAINING

- Google AI Professional Certificate, Google and Coursera, April 2026, seven course program with a portfolio of more than 20 AI artifacts.
- AI Fluency, Framework and Foundations, Anthropic with University College Cork and Ringling College, 2026.
- Python for Machine Learning, IBM.
- Python for Data Science, IBM.
- PyTorch and Deep Learning for Decision Makers, The Linux Foundation.
- McKinsey Forward Program, McKinsey dot org, December 2025.
- Registered Electrical Engineer, Pakistan Engineering Council, active since September 2021.

SKILLS

Applied AI and LLMs	LangChain, LlamaIndex, prompt engineering, retrieval augmented generation, AI agents, Model Context Protocol, embeddings, vector databases (ChromaDB, Pinecone), LLaMA via Groq, OpenAI, Gemini.
Machine learning	PyTorch based forecasting and generative models, scikit learn for clustering and classification, model evaluation with DeepEval.
Programming and data	Python, JavaScript, TypeScript, SQL, ETL pipeline design, data extraction and processing.
Deployment and tools	Docker (extensive), Kubernetes (foundational), Microsoft Azure, AWS, GitHub Actions, CI/CD, PostgreSQL, MySQL, MongoDB, FastAPI, n8n.
Responsible AI	Data privacy and PII redaction, model evaluation for hallucination and reliability, AI safety, ethics, and governance.
Languages	Urdu (mother tongue). English, C1 in listening, reading, writing, and spoken production; B2 in spoken interaction.

LEADERSHIP AND VOLUNTEERING

District Coordinator

Alkhidmat Foundation Pakistan

Volunteership | COVID 19 emergency response, and Pakistan floods emergency response, August to October 2025

- Coordinated district level emergency response across the four pillars of Rescue, Relief, Recovery, and Rehabilitation during the COVID 19 pandemic and the 2025 Pakistan floods.

- Organised and directed volunteer teams delivering rescue operations and emergency relief supplies to affected families under significant time pressure and limited resources.
- Coordinated recovery and rehabilitation efforts at the district level, working with local volunteers and community contacts to prioritise the most urgent needs.
- Applied the same structured, data informed coordination approach used in professional operations and quality assurance roles to a high pressure, resource constrained humanitarian setting.

HONOURS AND AWARDS

- High Performance Excellence Award, Allama Iqbal Hospital Kasur, June 2025.
- Productivity Leader Award, Ihsan Solar Energy Pvt. Ltd., July 2023.
- Top Innovative Final Year Project of the Year, IGNITE and Higher Education Commission, March 2021.